

Airbnb Dynamic Pricing Recommendation Engine

Python Data Analysis – Output Screenshots and Explanation

This PDF explains the Python output shown in the uploaded VS Code screenshots. The analysis is part of the Airbnb Dynamic Pricing Recommendation Engine project.

1. Dataset Information

The Python program successfully loaded the Airbnb dataset and displayed its structure. The screenshot shows 62 entries and 13 columns. The dataset contains information such as city, neighbourhood, property type, room type, number of guests accommodated, bedrooms, bathrooms, amenities, price, review score and number of reviews.

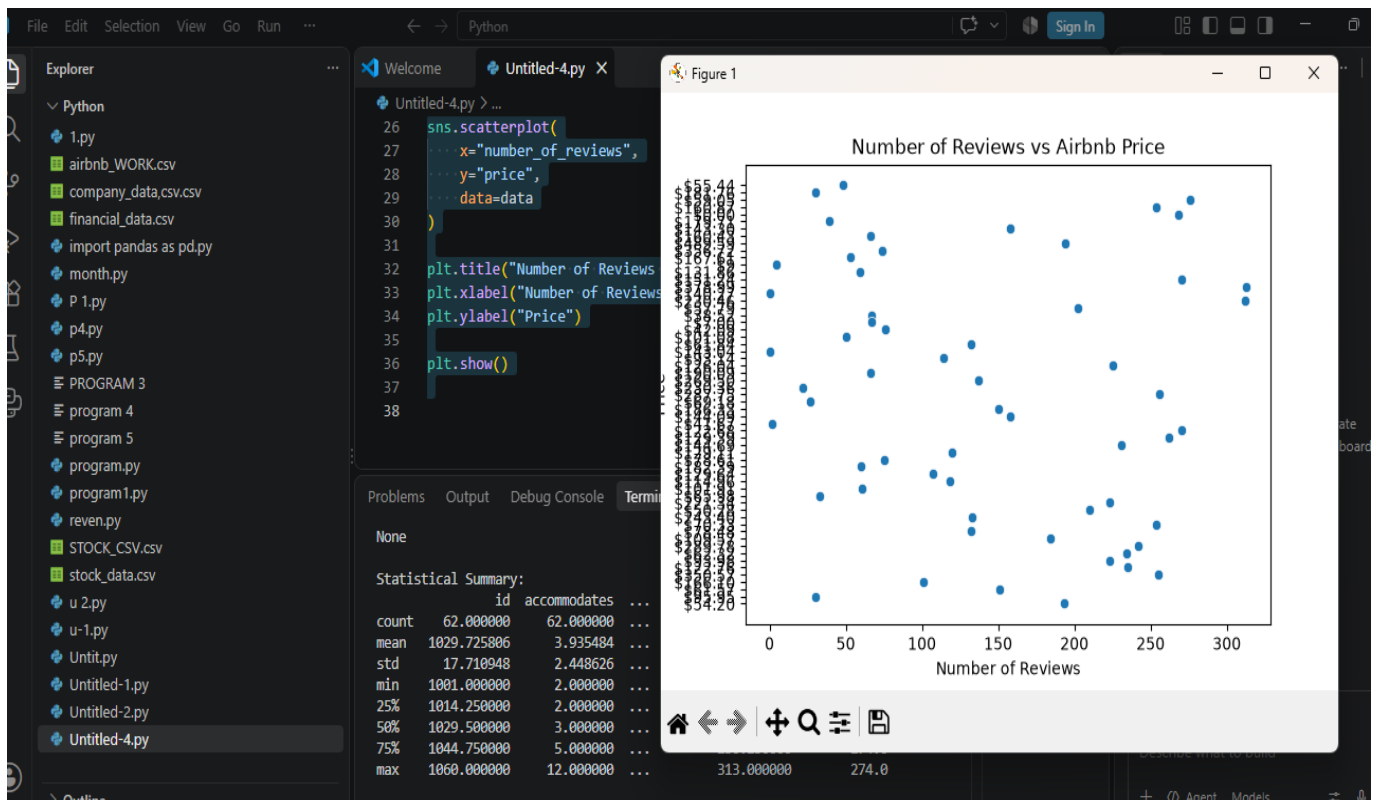


Figure 1: Python dataset information and statistical summary in VS Code.

2. Explanation of the Dataset Columns

Column	Explanation
id	Unique identification number for each Airbnb listing.
city	City where the Airbnb property is located.
neighbourhood	Specific neighbourhood or local area of the listing.
property_type	Type of property, such as an entire home or hotel-related property.
room_type	Type of accommodation offered to guests.
accommodates	Number of guests the property can accommodate.
bedrooms	Number of bedrooms in the property.
bathrooms	Number of bathrooms in the property.
amenities	Facilities provided by the property.

price	Listing price; this is the main target variable for price prediction.
review_scores_rating	Guest review rating, used as an indicator of listing quality.
number_of_reviews	Total number of reviews received by the listing.
Unnamed: 12	An extra/unused column detected in the dataset; it can normally be removed during cleaning.

3. Statistical Summary

The program also generated a statistical summary using `describe()`. For numerical columns, this summary shows count, mean, standard deviation, minimum, 25th percentile, median (50%), 75th percentile and maximum values.

Meaning of the statistics:

Statistic	Meaning
Count	Number of available observations.
Mean	Average value.
Std	Standard deviation, showing how spread out the values are.
Min	Smallest observed value.
25%	First quartile.
50%	Median value.
75%	Third quartile.
Max	Largest observed value.

4. Number of Reviews vs Airbnb Price

The second screenshot shows a scatter plot titled 'Number of Reviews vs Airbnb Price'. The horizontal axis represents the number of reviews and the vertical axis represents price. Each dot represents one Airbnb listing. The graph is useful for visually checking whether the number of reviews appears to have a clear relationship with price.

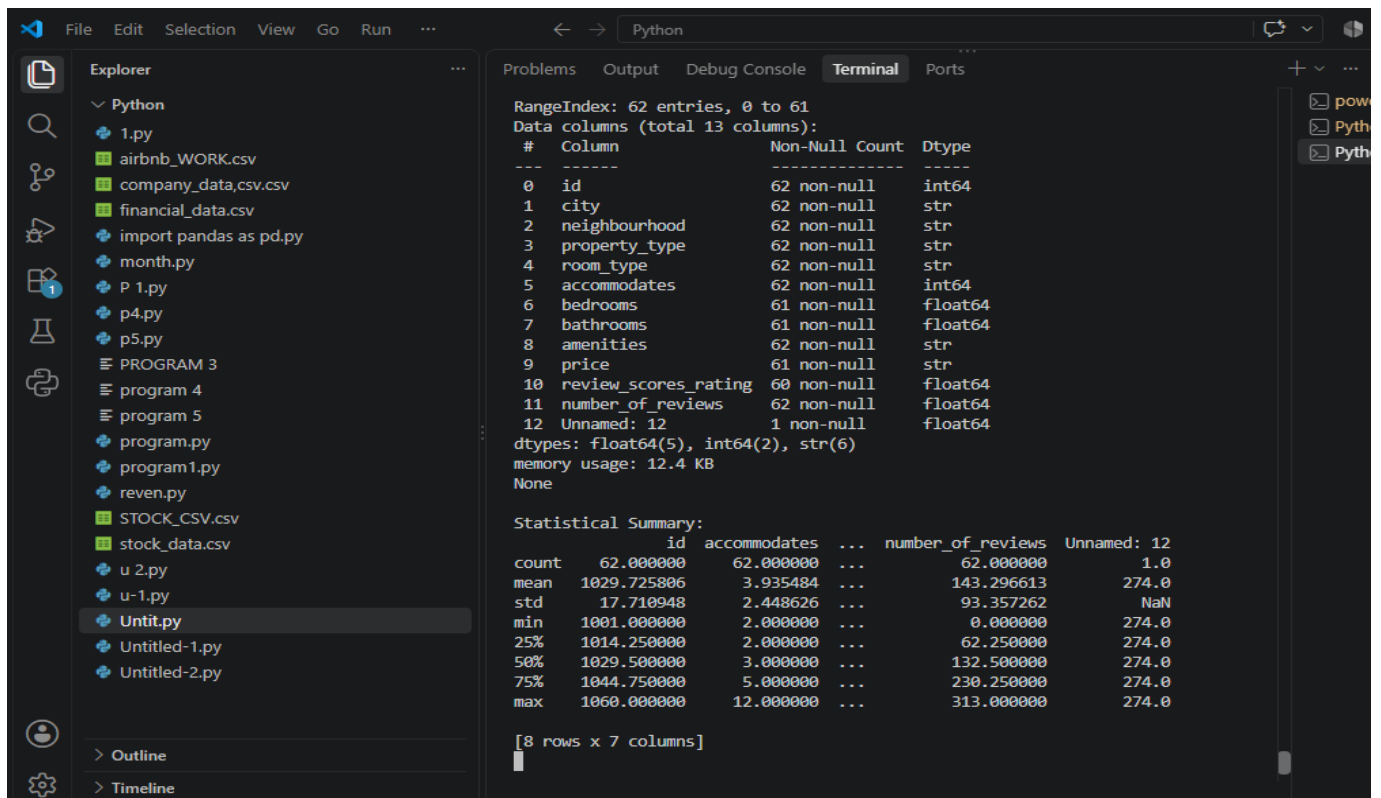


Figure 2: Scatter plot of number of reviews against Airbnb price.

5. Interpretation of the Scatter Plot

The points are spread across different review counts and price values rather than forming a single clear straight-line pattern. Therefore, number of reviews alone may not be sufficient to predict Airbnb price accurately. Other variables such as location, property type, room type, accommodation capacity, bedrooms, bathrooms and review score should also be considered.

6. Next Step: Regression Model

After exploratory analysis, the project can use Linear Regression to estimate Airbnb price. The price column is the dependent variable, while suitable listing characteristics are used as independent variables. The dataset is divided into training and testing data. The model is trained using the training data and evaluated using the testing data.

7. Important Dataset Cleaning Observation

The screenshot shows that some columns contain fewer non-null values than the total 62 rows. For example, bedrooms has 61 non-null values and review_scores_rating has 60 non-null values. Before regression, missing numerical values should be handled, for example by replacing them with the median or by removing affected rows. The 'Unnamed: 12' column also appears to be an extra column and can be removed if it contains no useful project information.

8. Project Flow

Step	Activity
1	Load Airbnb CSV into Python
2	Inspect columns and data types
3	Handle missing values and clean price
4	Analyze price by city and property type
5	Analyze reviews and other listing features
6	Build Linear Regression model
7	Evaluate prediction performance
8	Generate recommended price
9	Export predicted prices to CSV
10	Use the output in Tableau dashboard

Conclusion

The Python output confirms that the Airbnb dataset has been successfully loaded and explored. The statistical summary and scatter plot provide the initial exploratory analysis required before building the pricing model. The next stage is to clean the missing values, prepare the predictor variables, train the regression model and generate the final dynamic pricing recommendation.